

Full Stack Project Specification Document

1. Project Overview

1.1 Project Title

Business Management SaaS Platform - Multitenant System for Financial Operations and Team Collaboration

1.2 Project Description

A comprehensive Business Management SaaS Platform designed for businesses to manage their financial operations, invoices, expenses, team collaboration, and client relationships. The platform uses multitenant architecture.

2. Objectives

- Develop a multitenant SaaS platform for business management
- Implement core financial operations (invoicing, expense tracking)
- Enable team collaboration and client relationship management
- Provide reporting and analytics capabilities
- Ensure compliance with Tunisian data protection laws (data sovereignty)
- Create a scalable, production-ready platform

3. Scope

Project Phases

Phase 1:

- User authentication and authorization
- Business management
- Invoicing system
- Expense tracking
- Client management
- Basic reporting and analytics

Phase 2 (Production) - Optional:

- Enhanced features for commercial deployment
- Advanced reporting
- Performance optimization

Phase 3 (Future) - Beyond PFE Scope:

- Complex integrations
- Advanced features

4. Technical Requirements

4.1 Technology Stack

Frontend:

- React / Vue.js / Angular
- HTML5, CSS3

- UI Framework (Tailwind CSS / Material-UI / Bootstrap)
- Chart libraries for visualization

Backend:

- Node.js (Express / NestJS)
- RESTful API architecture
- JWT authentication
- PDF generation
- Email notifications

Architecture:

- Multitenant SaaS architecture
- Role-based access control (RBAC)
- Data encryption
- API documentation (Swagger)

4.2 Database**Options:**

- PostgreSQL (Recommended)
- MySQL
- MongoDB

Critical Requirement:

- **Data must be stored on servers physically located in Tunisia** (Data Sovereignty - Tunisian law compliance)

5. Functional Requirements**5.1 Actors**

1. **Platform Administrator** - Overall platform management
2. **Business Owner** - Full business control and configuration
3. **Business Administrator** - User and settings management
4. **Accountant** - Financial operations and reporting
5. **Team Member** - Limited access based on permissions
6. **Client (External)** - View invoices and make payments

5.2 Functionalities**5.2.1 Authentication & User Management**

- User registration and login
- Password reset
- Role-based access control
- User profile management

5.2.2 Multitenant Architecture

- Tenant creation and provisioning
- Data isolation between tenants
- Tenant-specific configuration

5.2.3 Business Management

- Business profile creation
- Multiple businesses per user
- Business settings and branding

- Tax and currency configuration

5.2.4 Invoicing System

- Invoice creation with line items
- Invoice templates
- Tax calculation
- Invoice status tracking
- PDF generation and email sending
- Payment tracking

5.2.5 Expense Management

- Expense entry and categorization
- Receipt upload
- Expense approval workflow
- Expense reporting

5.2.6 Client Management

- Client database
- Client contact information
- Client billing details
- Communication history

5.2.7 Reporting & Analytics

- Dashboard with key metrics
- Revenue and expense reports
- Profit/loss statements
- Data export (CSV, PDF)
- Visual charts and graphs

5.2.8 Team Collaboration

- Team member invitation
- Role and permission assignment
- Activity notifications
- Audit logs

6- Communication and Collaboration

Git , Github Action, Git Project

Logo